Contents

Moments

EEE 554 — Lecture #3

Mean

The first moment (mean) of a RV x:

$$\mu_x = E[x] = \int_{-\infty}^{\infty} x f_x(x) dx$$

may be interpreted physically as the center of mass of the PDF f_x. In other words, if one had a long bar of material whose density as a function of position x along the bar is given by $f_x(x)$, then μ_x is the point at which the bar would balance on a fulcrum.

Variance

The second central moment (variance) of x:

$$\sigma_x^2 = \text{var}(x) = E[(x - \mu_x)^2] = E[x^2] - \mu_x^2$$

can be interpreted as the moment of inertia around the mean.

Skewness

The skewness of a RV x with variance σ_x^2 and mean μ_x is:

$$r_x = \frac{E[(x - \mu_x)^3]}{\sigma_x^3}$$

Because $(x - \mu_x)^3 = x^3 - 3x^2\mu_x + 3x\mu_x^2 - \mu_x^3$, this can be expanded as:

$$r_x = \frac{E[x^3] - 3\mu_x E[x^2] + 2\mu_x^3}{\sigma_x^3}$$

Roughly speaking, skewness measures the asymmetry of the PDF around its mean μ_x. * **Positive skew:** Tail extends to the right. * **Negative skew:** Tail extends to the left.

Kurtosis

The (excess) kurtosis of a RV x with mean μ_x and variance σ_x^2 is:

$$K_x = \frac{E[(x-\mu_x)^4]}{\sigma_x^4} - 3$$

Expanding the expectation yields:

$$K_x = \frac{E[x^4] - 4\mu_x E[x^3] + 6\mu_x^2 E[x^2] - 3\mu_x^4}{\sigma_x^4} - 3$$

Markov and Chebyshev Inequalities

(Markov Inequality): Suppose a RV x is non-negative. Then for any $a > 0$:

$$P(x \geq a) \leq \frac{E[x]}{a}$$

Proof:

$$\begin{aligned} E[x] &= \int_0^\infty x f_x(x) dx \\ &= \int_0^a x f_x(x) dx + \int_a^\infty x f_x(x) dx \\ &\geq \int_a^\infty a f_x(x) dx \\ &= a P(x \geq a) \end{aligned}$$

(Chebyshev Inequality): If a RV x has mean μ_x and variance σ_x^2:

$$P(|x-\mu_x| > \epsilon) \leq \frac{\sigma_x^2}{\epsilon^2}$$

Proof:

$$\begin{aligned} \sigma_x^2 &= \int_{-\infty}^\infty (x-\mu_x)^2 f_x(x) dx \\ &\geq \int_{|x-\mu_x|>\epsilon} (x-\mu_x)^2 f_x(x) dx \\ &\geq \int_{|x-\mu_x|>\epsilon} \epsilon^2 f_x(x) dx \\ &= \epsilon^2 P(|x-\mu_x| > \epsilon) \end{aligned}$$

Examples of Common PDFs

Uniform RV: $x \sim U[a,b]$

Parameters: $a < b$

$$f_x(x) = \begin{cases} \frac{1}{b-a} & x \in [a,b] \\ 0 & \text{otherwise} \end{cases}$$

Moments:

$$E[x] = \int_a^b x \left(\frac{1}{b-a}\right) dx = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}$$

$$E[x^2] = \int_a^b x^2 \left(\frac{1}{b-a}\right) dx = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + ab + a^2}{3}$$

$$\therefore \text{var}(x) = E[x^2] - (E[x])^2 = \frac{(b-a)^2}{12}$$

Normal (Gaussian) RV: $x \sim N(\mu, \sigma^2)$

Parameters: $\mu \in \mathbb{R}$, $\sigma^2 > 0$

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Note: Define $I = \int_{-\infty}^{\infty} e^{-ax^2} dx$. Then with $a > 0$:

$$\begin{aligned}
\frac{I^2}{4} &= \int_0^\infty \int_0^\infty e^{-ax^2} e^{-ay^2} dx dy \\
&= \int_0^{\pi/2} \int_0^\infty e^{-ar^2} r \, dr \, d\theta \\
&= \frac{\pi}{2} \left(-\frac{e^{-ar^2}}{2a}\right)\Bigg|_0^\infty = \frac{\pi}{4a}
\end{aligned}$$

Thus, $I = \sqrt{\frac{\pi}{a}}$. In particular, substituting $a = \frac{1}{2\sigma^2}$, we get $\int_{-\infty}^{\infty} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx = \sqrt{2\pi\sigma^2}$, so that $\int_{-\infty}^{\infty} f_x(x) dx = 1$.

- **Mean:** Direct integration gives $E[x] = \mu$.
- **Variance:** Integration by parts gives $\text{var}(x) = \sigma^2$.

Exponential RV

Parameter: $\lambda > 0$

$$f_x(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & x < 0 \end{cases}$$

Characteristic Function:

$$\begin{aligned}
\Phi_x(\omega) &= \int_{-\infty}^{\infty} f_x(x) e^{-i\omega x} dx \\
&= \int_0^\infty \lambda e^{-(\lambda + i\omega)x} dx \\
&= \frac{\lambda}{\lambda + i\omega}
\end{aligned}$$

Notice that $\Phi_x(0) = 1 = \int_{-\infty}^{\infty} f_x(x) dx$.

Mean:

$$\Phi_x'(\omega) = \frac{-i\lambda}{(\lambda + i\omega)^2} \implies E[x] = \frac{\Phi_x'(0)}{-i} = \frac{1}{\lambda}$$

Variance:

$$\Phi_x''(\omega) = \frac{2(-i)^2\lambda}{(\lambda + i\omega)^3} \implies E[x^2] = \frac{\Phi_x''(0)}{(-i)^2} = \frac{2}{\lambda^2}$$

$$\therefore \mathrm{var}(x) = E[x^2] - (E[x])^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Higher Moments:

$$\Phi_x^{(n)}(\omega) = \frac{n!(-i)^n \lambda}{(\lambda + i\omega)^{n+1}} \implies E[x^n] = \frac{n!}{\lambda^n}$$

Using these moments to find skewness and kurtosis:

$$r_x = \lambda^3 \left[\frac{3!}{\lambda^3} - \frac{3}{\lambda} \cdot \frac{2}{\lambda^2} + \frac{2}{\lambda^3}\right] = 2$$

$$K_x = 6$$